subnetting basics

1. **Given the network 192.168.10.0/25, calculate the subnet mask, network address, broadcast address, and the range of usable IP addresses.**

ANS:

Network: 192.168.10.0/25

1. Subnet Mask

/25 means **25 network bits** and **7 host bits**.

Binary subnet mask:

11111111.11111111.11111111.10000000

So the subnet mask is:

255.255.255.128

2. Network Address

The given network is:

192.168.10.0/25

So the **Network Address = 192.168.10.0**

3. Broadcast Address

A /25 subnet contains:

2^7 = 128 total addresses.

The first subnet ranges from .0 to .127.

Therefore:

Broadcast Address = 192.168.10.127

4. Usable IP Address Range

Exclude the network address .0 and broadcast address .127.

Usable range = 192.168.10.1 – 192.168.10.126

Final Answer

Item	Answer
Network	192.168.10.0/25

Item	Answer
Subnet Mask	**255.255.255.128**
Network Address	**192.168.10.0**
Broadcast Address	**192.168.10.127**
Usable IP Range	**192.168.10.1 – 192.168.10.126**
Total Addresses	128
Usable Addresses	**126**

2. **Your company wants to split the 10.0.0.0/24 network into four equal subnets for different departments (HR, Sales, Tech, Admin). List the network address, broadcast address, and usable IP range for each subnet.**

ANS:

Subnetting 10.0.0.0/24 into 4 Equal Subnets
A /24 network has **256 addresses**.
To create **4 equal subnets**, borrow **2 host bits**:
/24 → /26
Each /26 subnet has:

- **64 total addresses**
- **62 usable IP addresses**
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Department	Network Address	Usable IP Range	Broadcast Address
HR	**10.0.0.0/26**	10.0.0.1 – 10.0.0.62	**10.0.0.63**
Sales	**10.0.0.64/26**	10.0.0.65 – 10.0.0.126	**10.0.0.127**
Tech	**10.0.0.128/26**	10.0.0.129 – 10.0.0.190	**10.0.0.191**

Department	Network Address	Usable IP Range	Broadcast Address
Admin	**10.0.0.192/26**	10.0.0.193 – 10.0.0.254	**10.0.0.255**

Final Answer
Subnet mask for all four: 255.255.255.192 (**/26**)
Each department gets **62 usable IP addresses**.

3. A startup needs 60 usable IP addresses for its team. What is the smallest CIDR subnet you can assign, and what are the network address, broadcast address, and usable IP range for this subnet?

ANS:

Requirement: 60 usable IP addresses
For IPv4:
Usable addresses = 2^host bits – 2

We need at least 60 usable addresses:

- 5 host bits → $2^5 - 2 =$ **30** ❌
- 6 host bits → $2^6 - 2 =$ **62** ✅

So, the smallest subnet is **/26**.
Assuming we assign the first /26 subnet from a private range:

Subnet: 192.168.1.0/26

Item	Answer
CIDR	192.168.1.0/26
Subnet Mask	255.255.255.192
Network Address	192.168.1.0
Broadcast Address	192.168.1.63
Usable IP Range	192.168.1.1 – 192.168.1.62

Item	Answer
Usable IPs	62

Answer: A **/26 subnet** is the smallest CIDR subnet that provides at least **60 usable IP addresses**.

**4. For the public WiFi at a Zomato restaurant, you are given the IP block 172.16.5.0/26. Calculate the subnet mask, network address, broadcast address, and the first and last usable IPs.

Hint: /26 gives you 64 addresses per subnet.**

ANS:

Given IP Block: 172.16.5.0/26
A /26 subnet has 64 total addresses.
Calculation:

- Total addresses = 2^(32−26) = 2^6 = 64
- Usable addresses = 64 − 2 = 62

Item	Answer
IP Block	172.16.5.0/26
Subnet Mask	255.255.255.192
Network Address	172.16.5.0
Broadcast Address	172.16.5.63
First Usable IP	172.16.5.1
Last Usable IP	172.16.5.62

Final Answer
Network: 172.16.5.0
Broadcast: 172.16.5.63
Usable IP range: 172.16.5.1 – 172.16.5.62
Subnet Mask: 255.255.255.192

5. You receive the following scenario: Flipkart's warehouse needs three subnets supporting at least 30 devices each. Using CIDR notation, assign suitable subnets from the 192.168.1.0/24 range, and specify the network address, broadcast address, and usable IP range for each subnet.

ANS:

Requirement

Flipkart's warehouse needs **3 subnets**, with **at least 30 usable IP addresses per subnet**, from 192.168.1.0/24.

For 30 devices:

- /27 gives 2^5 = 32 total addresses
- Usable = 32 − 2 = 30
- Therefore, **/27** is suitable.

Subnet Allocation

Subnet	CIDR	Network Address	Usable IP Range	Broadcast Address
Subnet 1	192.168.1.0/27	**192.168.1.0**	**192.168.1.1 – 192.168.1.30**	**192.168.1.31**
Subnet 2	192.168.1.32/27	**192.168.1.32**	**192.168.1.33 – 192.168.1.62**	**192.168.1.63**
Subnet 3	192.168.1.64/27	**192.168.1.64**	**192.168.1.65 –**	**192.168.1.95**

Subnet	CIDR	Network Address	Usable IP Range	Broadcast Address
			192.168.1.94	

Final Answer:

CIDR: /27
Subnet Mask: 255.255.255.224
Usable addresses per subnet: 30
The remaining range 192.168.1.96 – 192.168.1.255 remains available for future subnets/devices.